THE DIAGONAL OF THE PICARD FUNCTOR IS A QUASI-COMPACT IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1675-1678 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 1680-1689, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 30 of queue_moduli.json (pb=457, stmt 1675-1678, proof 1680-1689).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Editorial note. Immersion half is a named cross-chapter engine; the qc half is a displayed transfer argument.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DMA: sources/stacks-moduli/pinned-original-source.tex, source lines 1513–1516. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DME: sources/stacks-moduli/pinned-original-source.tex, source lines 1639–1642. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D2D: sources/stacks-moduli/pinned-original-source.tex, source lines 1689–1689. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DQL: sources/stacks-moduli/pinned-original-source.tex, source lines 1689–1689. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *The diagonal of* $\mathrm{Pic}_{X/B}$ *over* B *is a quasi-compact immersion.*

Proof. The diagonal is an immersion by Quot, Lemma 0D2D. To finish we show that the diagonal is quasi-compact. The diagonal of $\mathcal{P}ic_{X/B}$ is quasi-compact by Lemma 0DMA and $\mathcal{P}ic_{X/B}$ is a gerbe over $\mathrm{Pic}_{X/B}$ by Lemma 0DME. We conclude by Morphisms of Stacks, Lemma 0DQL. □